Day 3 of 15 · Student Activity Sheets

Day 3 Activities — Probability, Correlation, Regression and the t-Test

TESDA Data Analytics Level III · Batch DAL3-2026

Name: _____ Date: _____ Workstation #: _____

File you need: `Day03/assets/Day03_Analysis_Template.xlsx`

Before you begin:

1. Copy the file into `DAL3/Day03/`, open it, and **Save As** `Day03_<Surname>.xlsx`.
2. Look at the sheets (**Ctrl + Page Down** moves to the next sheet): **Probability**, **Rice_Weights**, **Temp_Drinks**, **Price_Qty**, **Branch_Test**, **Promo_Test**.
3. **Yellow cells** = type your formula here. **Green cell** = a value given to you.
4. Stuck? Re-watch that chapter of the demo video. The PRESS box shows every key the trainer pressed.

Activity	Title	Mode	Time	Points
3.0	Warm-up: Compute by Hand	Individual	20 min	10
3.1	Probability of Payment Methods	Individual	25 min	15
3.2	z-Scores and the Normal Curve (rice bags)	Individual	35 min	20
3.3	Correlation and Scatter Charts	Individual	30 min	15
3.4	Regression Forecast	Individual	30 min	15
3.5	t-Test: Real Difference or Chance?	Pair	30 min	15
3.6	Recommendations for Aling Nena	Group of 3–4	25 min	10
	Total		**≈ 3 hrs 15 min**	**100**

Activity 3.0 — Warm-up: Compute by Hand (10 pts)

Use a calculator, not Excel. Show your solution.

a) Probability. A sari-sari store's jar has 8 Choc-Nut, 5 Haw Flakes and 7 White Rabbit candies. One candy is picked without looking.

Question	Solution	Answer
P(Choc-Nut)		
P(not Choc-Nut)		

Question	Solution	Answer
P(Haw Flakes or White Rabbit)		

b) z-score. A class quiz has mean = 30 and SD = 4. Compute the z-score of each score and say whether it is *normal* or *unusual*.

Score	z = (score − 30) ÷ 4	Normal or unusual?
34		
21		
39		

c) Read the scatter. For each description, write *strong positive*, *strong negative*, or *none*:

1. Hours of brownout vs ice sold at the store: _____
2. Price of rice per kilo vs kilos sold: _____
3. Customer's shoe size vs their GCash balance: _____

Activity 3.1 — Probability of Payment Methods (15 pts)

Sheet **Probability** (500 transactions from Day 1).

1. **G4:** `=COUNTIF(C2` → press **F4** → `:C501` → press **F4** → `,F4)` → **Enter**. The result should be `=COUNTIF($C$2:$C$501,F4)` (cell F4 holds the word *Cash*).
2. Select **G4:G7** (click G4, then **Shift** + ↓ three times) → **Ctrl + D**.
3. **G8:** press **Alt + =** → **Enter**. It must show **500**.
4. **H4:** `=G4/G8` → press **F4** so G8 becomes `$G$8` → **Enter** → fill down to H7. **H8:** Alt + =.
5. **I4:** `=H4` → fill down to I7, **I8:** Alt + = → select **I4:I8** → **Ctrl + Shift + %**.
6. **G10:** `=COUNTA(C2:C501)`. **G12:** P(not Cash) `=1-H4`. **G13:** P(GCash or Maya) `=H5+H6`. **G14:** expected GCash payers in the next 200 customers `=200*H5`.

Questions

a) What is the most likely payment method? _____ Its probability: _____

b) Explain P(GCash) in one plain sentence that Aling Nena would understand.

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c) Why did we lock `$G$8` but not `G4` in step 4?

Activity 3.2 — z-Scores and the Normal Curve (20 pts)

Sheet **Rice_Weights**: 80 bags labelled **5 kg**.

1. **G3** mean `=AVERAGE(B2:B81)` · **G4** SD `=STDEV.S(B2:B81)` · **G5** `=COUNT(B2:B81)`.
2. **C2** z-score: `=(B2-G3` → **F4** → `)/G4` → **F4** → **Enter** → result `=(B2-$G$3)/$G$4`.
3. Copy it to all 80 bags: click **C2** and **double-click the fill handle** (small square at the bottom-right of the cell).
4. **D2:** `=IF(ABS(C2)>2,"Unusual","")` → double-click the fill handle.
5. **G7:** `=COUNTIF(D2:D81,"Unusual")`.
6. **G9:** `=NORM.DIST(5,G3,G4,TRUE)` → **Ctrl + Shift + %**. **G10:** `=G9*1000`.
7. **G12:** % of bags within ±1 SD `=COUNTIFS(B2:B81,">="&G3-G4,B2:B81,"<="&G3+G4)/G5`. **G13:** the same for ±2 SD (use `2*G4`).
8. **G14:** `=STANDARDIZE(4.95,G3,G4)`.

Questions

a) List the unusual bags (bag number and z-score):

Bag No.	Weight (kg)	z-score	Under- or over-filled?

b) Compare G12 and G13 with the empirical rule (68% and 95%). Is the bag weight approximately normal?

_____

c) G9 is about 16%. What should Aling Nena do about the packing scale, and why does it matter to customers? *(Integrated: Apply critical thinking & problem-solving.)*

Activity 3.3 — Correlation and Scatter Charts (15 pts)

Part A, sheet Temp_Drinks (60 days, Quezon City)

1. **G3:** `=CORREL(B2:B61,C2:C61)`.
2. Scatter chart: click **B1** → **Ctrl + Shift +** ↓ → **Shift +** → → **Insert** → **Scatter (first icon)**. Move the chart beside the table.
3. Add chart title *Hotter days, more cold drinks*, and axis titles *Max temperature (°C)* and *Cold drinks sold* (**Chart Design** → **Add Chart Element** → **Axis Titles**).

Part B, sheet Price_Qty (20 weeks of Lucky Me Pancit Canton)

4. **G3:** `=CORREL(B2:B21,C2:C21)`. Make a scatter chart of Price (x) vs Packs_Sold (y).

	r	Direction	Strength
Temperature vs cold drinks			
Price vs packs sold			

a) Write one sentence explaining each r in plain words.

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b) Ice-cream sales and drowning cases are positively correlated. Does ice cream cause drowning? Explain using the word *causation*.

Activity 3.4 — Regression Forecast (15 pts)

Sheet **Temp_Drinks**.

1. Add a trendline: select the chart → **Chart Design** → **Add Chart Element** → **Trendline** → **Linear**. Then right-click the line → **Format Trendline** → ✓ *Display Equation on chart* and ✓ *Display R-squared value*.
2. **G4** `=RSQ(C2:C61,B2:B61)` · **G5** `=SLOPE(C2:C61,B2:B61)` · **G6** `=INTERCEPT(C2:C61,B2:B61)`. Remember: **y range first**.
3. **G9:** `=FORECAST.LINEAR(G8,C2:C61,B2:B61)` (G8 = 37 °C, green cell). **G10** check: `=G6+G5*G8`. G9 and G10 must match.
4. Sheet **Price_Qty**: G4 slope, G5 intercept, G6 forecast at ₱15.00.
5. *(Optional, ToolPak)* **Data** → **Data Analysis** → **Regression** → Input Y `$C$1:$C$61`, Input X `$B$1:$B$61`, ✓ Labels, New Worksheet Ply. Find *R Square* and the *Coefficients*. Do they match G4–G6?

Item	Temp_Drinks	Price_Qty
Slope		
Intercept		
R^2		—
Forecast	at 37 °C:	at ₱15:

a) Explain the slope of Temp_Drinks in plain words ("For every…").

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b) Change G8 to 45 °C. Should Aling Nena trust that forecast? Why or why not?

Activity 3.5 — t-Test: Real Difference or Chance? (Pair, 15 pts)

Sheet Branch_Test (Marikina vs Pasig, 90 days)

1. **G4** `=AVERAGE(B2:B91)` → select **G4:H4** → **Ctrl + R** (fill right). Do the same for G5 (`=STDEV.S(B2:B91)`) and G6 (`=COUNT(B2:B91)`).

2. **G8** `=H4-G4` · **G9** `=T.TEST(B2:B91,C2:C91,2,3)` · **G10** `=IF(G9<0.05,"YES - real difference","NO - could be chance")`.
3. Cross-check: **Data** → **Data Analysis** → **t-Test: Two-Sample Assuming Unequal Variances** → Variable 1 `$B$1:$B$91`, Variable 2 `$C$1:$C$91`, ✓ Labels, New Worksheet Ply `t-Test` → OK. Read **P(T<=t) two-tail**.

Sheet Promo_Test (Caloocan basket size, 25 days before vs 25 days after the new shelf layout)

4. Repeat steps 1–2 with ranges **B2:B26** and **C2:C26**.

Test	Mean 1	Mean 2	p-value	Significant?
Marikina vs Pasig				
Before vs after layout				
ToolPak P two-tail (Branch_Test)				Matches G9? ✓/✗

a) Write H_0 and H_1 for the Branch_Test in your own words.

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b) The new layout raised the average basket by about ₱3, but p ≈ 0.27. What should you tell Aling Nena? What would you do next?

Activity 3.6 — Recommendations for Aling Nena (Group, 10 pts)

(Integrated: Apply critical thinking & problem-solving; Propose learning & innovation.)
As a team, write **four recommendations**, one for each result. Use *What* → *So what* → *Now what*:

1. Payment methods (Activity 3.1)
2. Rice-bag weights (Activity 3.2)
3. Hot-weather stocking (Activities 3.3–3.4)
4. Branches and the new layout (Activity 3.5)

Rubric: correct numbers (4), plain-language explanation (3), practical action (3). The reporter presents in 2 minutes.

Daily Reflection Log

1. The new term I can now explain to a friend, in my own words:

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2. The shortcut key I will use from now on:

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3. Where could I use a forecast or a t-test at work or at home?

Submit: Day03_<Surname>.xlsx (all six sheets, the scatter charts with trendline, and the t-Test sheet) and this sheet.